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EXP28:

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#include <stdio.h>

typedef unsigned long long u64;

u64 modpow(u64 b, u64 e, u64 m) {
    u64 r = 1; b %= m;
    while (e) { if (e & 1) r = r * b % m; b = b * b % m; e >>= 1; }
    return r;
}
long long egcd(long long a, long long b, long long *x, long long *y) {
    long long x1, y1, g;
    if (b == 0) { *x = 1; *y = 0; return a; }
    g = egcd(b, a % b, &x1, &y1);
    *x = y1; *y = x1 - (a / b) * y1;
    return g;
}

int main() {
    u64 q = 353, a = 3, xa, xb, ya, yb, ka, kb;
    long long ainv, t;

    printf("Public values: q = %llu, a = %llu\n", q, a);
    printf("Alice's secret xA: ");
    scanf("%llu", &xa);
    printf("Bob's secret xB  : ");
    scanf("%llu", &xb);

    /* Standard Diffie-Hellman: send a^x mod q */
```

```
    ya = modpow(a, xa, q);
    yb = modpow(a, xb, q);
    ka = modpow(yb, xa, q);
    kb = modpow(ya, xb, q);
    printf("\n--- Standard Diffie-Hellman (a^x mod q) ---\n");
    printf("Alice sends YA = %llu, Bob sends YB = %llu\n", ya, yb);
    printf("Alice's key = %llu, Bob's key = %llu -> %s\n", ka, kb, ka == kb ? "keys agree" :
"keys differ");
    printf("Eve sees only YA, YB: she must solve a discrete logarithm (hard).\n");

    /* Modified protocol: send x^a mod q */
    ya = modpow(xa, a, q);
    yb = modpow(xb, a, q);
    printf("\n--- Modified protocol (x^a mod q) ---\n");
    printf("Alice sends %llu, Bob sends %llu\n", ya, yb);
    egcd(a, q - 1, &ainv, &t);
    ainv = ((ainv % (long long)(q - 1)) + (q - 1)) % (q - 1);
    printf("Eve computes a^-1 mod (q-1) = %lld and takes the a-th root:\n", ainv);
    printf("  xA = %llu^%lld mod q = %llu\n", ya, ainv, modpow(ya, ainv, q));
    printf("  xB = %llu^%lld mod q = %llu\n", yb, ainv, modpow(yb, ainv, q));
    printf("Eve finds the secret numbers easily, so she can compute any key\n");
    printf("(e.g. K = xA^xB mod q = %llu). The modified protocol is insecure.\n",
        modpow(modpow(ya, ainv, q), modpow(yb, ainv, q), q));
    return 0;
}
```

OUTPUT:

```
Public values: q = 353, a = 3
Alice's secret xA: 97
Bob's secret xB  : 233

--- Standard Diffie-Hellman (a^x mod q) ---
Alice sends YA = 40, Bob sends YB = 248
Alice's key = 160, Bob's key = 160 -> keys agree
Eve sees only YA, YB: she must solve a discrete logarithm (hard).

--- Modified protocol (x^a mod q) ---
Alice sends 168, Bob sends 288
Eve computes a^-1 mod (q-1) = 235 and takes the a-th root:
  xA = 168^235 mod q = 97
  xB = 288^235 mod q = 233
Eve finds the secret numbers easily, so she can compute any key
(e.g. K = xA^xB mod q = 122). The modified protocol is insecure.
```